INCH-POUND
MIL-DTL-24643/53E
22 August 2002
SUPERSEDING
MIL-C-24643/53D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 3000 VOLTS, TYPES LSSRW, LSDRW AND LSTRW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, uncoated, AWG size No. 14, class B stranding.
Second -	Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Cross-linked polyethylene insulation. Insulation wall thickness 0.080 inch minimum average. Standard identification code shall be applied by method 1 or method 3. The insulation shall be colored black if method 1 is used.
Fourth -	The required two or three conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.
Fifth -	Binder tape applied helically with overlap for two and three conductor cables only.
Sixth -	Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

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TABLE I. Details.

Military part no. M24643/53	Cable Type	Number of Conductors	Overall diameter maximum (inch)	Conductor resistance (ohms)	Cold bending mandrel diameter (max inch)
-01UN	LSSRW	1	0.400	2.69	3
-02UN	LSDRW	2	0.670	2.75	5
-03UN	LSTRW	3	0.710	2.75	7

EXAMINATION AND TESTS:

Basic electrical: Requirements

Conductor Resistance - ohms/1000 feet at 25°C, maximum (see table I)
Voltage withstand - volts, root mean square, minimum 8000
Insulation resistance - Conductor to conductor,
Megohms -1000 feet, minimum 9500
Conductor continuity ..... No failure

Group A:
Visual and dimensional ..... No failure
Watertightness - see MIL-DTL-24643 for water leakage ..... No failure

Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C) 50
Cold bending, cable - (see table I for mandrel diameter) ..... No damage
Drip - 95 ± 1°C ..... Zero
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80

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Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (conductor) - (method 1 only) cycles, minimum (LSDRW and LSTRW) 25
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability ..... No failure

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80

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Weathering (jacket ..... No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSSRW	2500
LSDRW and LSTRW	2000

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-051)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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